

NEUQUEN, Argentina

WMO: 877150

Lat: 38.952S

Lon: 68.137W

Elev: 271

StdP: 98.11

Time Zone: -3.00 (W03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	-3.7	-2.1	-10.8	1.5	13.4	-8.8	1.8	11.9	12.4	14.8	10.3	13.9	1.0	270	0.474

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	14.6	35.2	18.9	33.7	18.1	32.0	17.4	21.4	30.8	20.1	29.6	19.1	28.6	3.8	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.2	13.5	25.6	16.9	12.4	24.4	15.5	11.4	23.0	63.4	30.8	59.0	29.7	55.2	28.6	28.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.9	9.0	7.7	DB	-6.2	38.6	1.7	1.3	-7.5	39.5	-8.5	40.3	-9.5	41.0	-10.7	42.0	(4)
(5)			WB	-6.8	23.5	1.9	1.9	-8.2	24.9	-9.3	26.0	-10.3	27.1	-11.7	28.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	15.2	24.0	22.6	19.3	14.4	10.2	7.4	6.8	9.0	12.0	15.6	19.3	22.5	(6)	
	DBStd	6.77	3.03	3.33	3.38	3.41	3.24	3.17	3.10	3.37	3.22	3.31	3.20	3.21	(7)	
	HDD10.0	314	0	0	0	4	37	89	107	59	16	2	0	0	(8)	
	HDD18.3	1704	2	4	29	123	253	328	357	289	192	96	26	5	(9)	
	CDD10.0	2220	433	353	287	137	43	12	8	29	75	176	281	386	(10)	
	CDD18.3	568	176	125	58	7	1	0	0	0	1	12	56	133	(11)	
	CDH23.3	6651	2044	1408	672	109	7	1	0	9	33	188	642	1538	(12)	
(13)	CDH26.7	2799	984	635	238	19	1	0	0	1	5	41	204	671	(13)	
(14)	Wind	WSAvg	2.9	3.5	3.1	2.7	2.4	2.1	2.4	2.4	2.7	3.1	3.4	3.6	3.8	(14)
(15) Precipitation	PrecAvg	219	15	16	19	25	26	17	14	11	16	35	16	11	(15)	
	PrecMax	462	54	88	113	230	87	61	39	49	60	198	90	55	(16)	
	PrecMin	82	0	0	1	1	0	0	0	0	0	0	0	1	(17)	
	PrecStd	108	14	21	24	44	27	18	12	12	16	42	22	15	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	38.0	36.8	34.4	28.9	24.0	20.6	20.1	24.4	26.8	30.2	33.8	36.7	(19)	
		MCWB	20.2	20.7	18.4	16.6	13.6	11.2	10.8	12.4	13.7	15.7	17.2	19.1	(20)	
	2%	DB	35.2	34.5	31.6	26.4	20.8	17.6	17.1	20.7	23.6	27.2	31.0	34.2	(21)	
		MCWB	18.7	19.3	17.9	15.3	12.3	9.8	9.3	11.0	12.2	14.0	16.3	18.0	(22)	
	5%	DB	33.7	32.8	29.5	24.0	18.4	15.6	15.2	18.0	21.4	25.2	29.0	32.2	(23)	
		MCWB	18.1	18.6	17.5	14.2	11.4	8.9	8.4	9.6	11.0	13.0	15.3	17.0	(24)	
	10%	DB	31.9	30.8	27.1	21.9	16.8	13.9	13.3	15.9	19.5	23.2	27.1	30.6	(25)	
		MCWB	17.1	17.8	16.3	13.3	10.6	8.2	7.4	8.5	10.1	12.2	14.3	16.5	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	22.8	23.8	21.7	18.0	15.2	12.3	11.8	13.7	15.3	17.3	19.7	22.0	(27)	
		MCDB	33.3	32.6	29.6	25.6	19.9	18.2	17.7	22.2	23.4	27.1	30.1	32.4	(28)	
	2%	WB	20.9	21.6	19.9	16.5	13.4	10.6	10.1	11.4	13.2	15.6	17.5	19.9	(29)	
		MCDB	31.2	30.5	27.7	23.2	18.6	15.8	15.8	18.7	21.5	24.1	28.0	30.4	(30)	
	5%	WB	19.6	20.2	18.5	15.3	12.3	9.6	9.1	10.2	11.9	14.3	16.3	18.5	(31)	
		MCDB	29.8	29.4	26.4	21.6	17.3	14.1	14.3	16.7	19.5	22.7	26.4	29.1	(32)	
	10%	WB	18.4	19.0	17.4	14.2	11.3	8.7	7.9	9.2	10.8	13.1	15.1	17.3	(33)	
		MCDB	28.9	28.3	25.4	20.2	15.6	12.8	12.4	14.9	18.0	21.2	25.3	28.2	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	14.6	14.4	14.2	13.4	11.4	10.9	11.8	12.7	13.1	13.0	13.8	14.3	(35)	
		MCDBR	17.3	17.3	17.7	16.8	14.4	13.4	14.8	16.0	16.8	16.4	16.5	16.8	(36)	
		MCWBR	6.5	6.6	7.3	7.9	7.9	7.6	8.6	8.4	8.3	7.0	6.7	6.6	(37)	
	5% WB	MCDBR	15.8	15.4	15.2	14.5	12.7	11.8	13.6	14.6	15.0	14.9	15.2	15.8	(38)	
		MCWBR	7.1	7.0	7.3	7.5	7.6	7.4	8.4	8.2	7.9	7.1	7.1	7.2	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.382	0.375	0.355	0.347	0.326	0.305	0.304	0.320	0.350	0.354	0.363	0.374	(40)	
	taud		2.321	2.353	2.415	2.418	2.469	2.540	2.536	2.477	2.375	2.367	2.352	2.342	(41)	
	Ebn at Noon		948	929	906	845	800	796	823	865	896	940	961	962	(42)	
	Edh at Noon		134	124	107	93	76	66	70	86	110	121	130	133	(43)	
(44) All-Sky Solar Radiation	RadAvg		8.45	7.22	5.63	3.80	2.35	1.92	2.17	3.06	4.55	6.06	7.73	8.57	(44)	
	RadStd		0.34	0.28	0.28	0.33	0.28	0.21	0.22	0.26	0.32	0.37	0.33	0.34	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.28	N/A	N/A	+0.56	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page